# Inductive Bias Meta-Learning with Generative Models

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## Abstract

This paper investigates inductive bias in machine learning models. By inductive bias we mean the preference of a model for certain types of functions or data structures over others. To analyze the inductive bias of a fixed model, we consider a problem of finding data that this model can fit and generalize on particularly well. Previous work demonstrated that generating labels for a fixed dataset allows one to extract the inductive bias. Here, we extend this approach by proposing a method for generating full synthetic datasets. We train a generative model to produce datasets on which the target model achieves strong generalization performance. We test the proposed framework on CNNs and RNNs, and analyze obtained datasets for each model.

**Keywords:** inductive bias, meta-learning, generative models.

## 1 Introduction

Inductive bias plays a fundamental role in a model's ability to generalize [1]. Without additional assumptions, any hypothesis that fits a finite training dataset is considered equally likely by the model, which can lead to poor generalization. Thus, effective generalization requires a set of prior assumptions about the data, and these assumptions constitute the model's inductive bias. Understanding and exploiting inductive bias can be a powerful tool for enhancing model performance.

The term "inductive bias" does not have a single, strict definition. For instance, inductive bias can mean underlying assumption about the domain and task or specific implementation choices that constrain and/or order the space of hypotheses [2]. In this paper, by inductive bias we mean a model's preference for certain types of functions or data structures over others, expressed through better generalization performance. For example, convolutional neural networks (CNNs) are inductively biased towards data with local structure and translation invariance [3–5], while recurrent neural networks (RNNs) are inductively biased towards ordered data with sequential structure [6–8].

A clear understanding of the correspondence between models (or their specific components) and inductive biases is useful for solving machine learning problems. However, this task is highly non-trivial: for most models with complex architecture, the problem of detecting inductive bias is analytically intractable. Several approaches exist for extracting and analyzing the inductive bias of models. Boopathy et al. [9] proposed a method to quantify the inductive bias of tasks or models as a prior probability mass required for accurate generalization at target risk, calculated via empirical sampling and parametric tail modeling. Si et al. [10] investigated the inductive bias of in-context learning by constructing underspecified demonstrations. Li et al. [11] propose a framework to capture the inductive biases in a learning system by meta-learning Gaussian process kernel hyperparameters from its predictions

One of recent results in this area is the meta-learning framework introduced by Dorrell et al. [12], which identifies the inductive bias of differentiable supervised learning algorithms by generating

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labels for a fixed dataset. For all its strengths, this approach has certain limitations. First, it requires access to input data or its distribution. Second, its analysis of inductive bias is constrained by the fixed dataset and its structure.

Here, we propose to extend this approach by generating entire synthetic datasets. The meta-learning loop proceeds as follows: an outer generative model, or meta-learner, produces a dataset, which is used to train and test an inner model, or learner, whose inductive bias we aim to extract. After the learner is trained, the quality of its predictions on test data from this dataset is incorporated into the meta-learner's loss function. Thus, the meta-learner learns to generate datasets on which the target model shows the best generalization performance; these are datasets towards which it is inductively biased.

The proposed approach requires no knowledge of the input data distribution, making it more flexible. We expect it to enable a more comprehensive analysis of inductive bias.

**Contributions.** Our contributions can be summarized as follows:

- We present a generative meta-learning framework to extract the inductive bias of a target model. Unlike previous work that only generates label for a samples from fixed distribution, our approach generates entire synthetic datasets via meta-learning generative model.
- We empirically evaluate our algorithm on CNNs and RNNs and compare the results with known inductive biases of these architectures. For CNNs, we expect to see locality in the structure of data in the resulting datasets. For RNNs, we expect to see an ordered sequential structure.
- We highlight the implications of our findings for understanding and leveraging inductive bias.

**Outline.** The rest of the paper is organized as follows...

## 2 Problem statement

We study the problem of extracting and characterizing the inductive bias of learning models by finding the datasets that they generalize best. We now present a formal statement of the problem.

Let $\mathbb{X} \subset \mathbb{R}^n$ be the input space, $\mathbb{Y} \subset \mathbb{R}^m$ be the output space (e.g., class labels) and $\mathbf{f}_{\boldsymbol{\theta}} : \mathbb{X} \to \mathbb{Y}$ be the target model, where $\boldsymbol{\theta} \in \Theta \subset \mathbb{R}^k$ are its parameters. Let $\ell(\mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}), \mathbf{y})$ be the loss function of the target model.

Consider a parametric family of distributions $p(\mathbf{x}|\mathbf{y}, \mathbf{w})$, where $\mathbf{w} \in \mathbb{W} \subset \mathbb{R}^l$ are the parameters of the distribution. For a fixed label $\mathbf{y} \in \mathbb{Y}$, a sample $\mathbf{x} \sim p(\mathbf{x}|\mathbf{y}, \mathbf{w})$ is generated from this distribution by the generator $g(\mathbf{w})$.

We aim to find data distribution function $p^*$, corresponding to parameters $\mathbf{w}^*$, the solution to the following optimization problem:

$$\mathbf{w}^* = \arg\min_{\mathbf{w} \in \mathbb{W}} \mathrm{E}_{\mathfrak{D}_{\text{train}}, \mathfrak{D}_{\text{test}} \sim p(\mathbf{x}|\mathbf{y},\mathbf{w})} \left[ \mathcal{L}\left(\mathbf{f}_{\hat{\boldsymbol{\theta}}}, \mathfrak{D}_{\text{test}}\right) \right] \quad \text{s.t.} \quad |\mathfrak{D}_{\text{train}} \cup \mathfrak{D}_{\text{test}}| = d$$

where

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta} \in \Theta} \sum_{(\mathbf{x},\mathbf{y}) \in \mathfrak{D}_{\text{train}}} \ell(\mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}), \mathbf{y}).$$

Function $\mathcal{L}$ measures a model's ability to generalize by calculating the prediction error on a test set and also contains regularization terms to avoid trivial solutions. For example, prediction error can be estimated as cross-entropy or mean squared error (MSE), defined respectively as:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{|\mathfrak{D}_{\text{test}}|} \sum_{(\mathbf{x},\mathbf{y}) \in \mathfrak{D}_{\text{test}}} \sum_{i=1}^{m} y_i \log\left([\mathbf{f}_{\hat{\boldsymbol{\theta}}}(\mathbf{x})]_i\right), \qquad \mathcal{L}_{\text{MSE}} = \frac{1}{|\mathfrak{D}_{\text{test}}|} \sum_{(\mathbf{x},\mathbf{y}) \in \mathfrak{D}_{\text{test}}} \|\mathbf{y} - \mathbf{f}_{\hat{\boldsymbol{\theta}}}(\mathbf{x})\|^2.$$

The regularization term can be the entropy of the found distribution, defined as:

$$\mathcal{R}_{\text{ent}}(\mathbf{w}) = -\mathrm{E}_{\mathbf{y}} \left[ \int_{\mathbb{X}} p(\mathbf{x}|\mathbf{y}, \mathbf{w}) \log p(\mathbf{x}|\mathbf{y}, \mathbf{w}) \, d\mathbf{x} \right].$$

# 3 Method

# 4 Experiments

To verify the theoretical estimates obtained, we conducted a detailed empirical study...

# 5 Discussion

TODO

# 6 Conclusion

TODO

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# A Appendix / supplemental material

## A.1 Additional experiments / Proofs of Theorems

TODO